Safety Data Sheet

According to Annex II to REACH - Regulation (EU) 2020/878 and to Annex II to UK REACH

SECTION 1. Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Code:	**SBPUS348GHL7011**
Product name	**ICROPUR TOP GLOSS 2K R7011 (LD)**
UFI :	**CV1U-20Q2-000U-PN1H**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Identified Uses	Industrial	Professional	Consumer
Paint product	✓	-	-

1.3. Details of the supplier of the safety data sheet

Name	**ICRO COATINGS S.P.A. CON SOCIO UNICO**		
Full address	**Via Bedeschi, 25**		
District and Country	**24040**	**Chignolo D'Isola**	**(BG)**
		Italia	
	Tel.	**+39 035 999711**	
	Fax	**+39 035 999712**	
e-mail address of the competent person responsible for the Safety Data Sheet	**gianluca.cerina@icro.it**		
Supplier:	**ICRO COATINGS S.p.A. con Socio Unico - Via Bedeschi 25 - 24040 Chignolo d'Isola (BG) - Italy**		

1.4. Emergency telephone number

For urgent inquiries refer to **There is no known operational emrgency phone number service in UK**

SECTION 2. Hazards identification

2.1. Classification of the substance or mixture

The product is classified as hazardous pursuant to the provisions set forth in (EC) Regulation 1272/2008 (CLP) (and subsequent amendments and supplements). The product thus requires a safety datasheet that complies with the provisions of (EU) Regulation 2020/878.
Any additional information concerning the risks for health and/or the environment are given in sections 11 and 12 of this sheet.

Hazard classification and indication:

Flammable liquid, category 2	H225	Highly flammable liquid and vapour.
Aspiration hazard, category 1	H304	May be fatal if swallowed and enters airways.
Specific target organ toxicity - repeated exposure, category 2	H373	May cause damage to organs through prolonged or repeated exposure.
Skin irritation, category 2	H315	Causes skin irritation.

2.2. Label elements

Hazard labelling pursuant to EC Regulation 1272/2008 (CLP) and subsequent amendments and supplements.

Hazard pictograms:



Signal words: Danger

SECTION 2. Hazards identification ... / >>

Hazard statements:

H225	Highly flammable liquid and vapour.
H304	May be fatal if swallowed and enters airways.
H373	May cause damage to organs through prolonged or repeated exposure.
H315	Causes skin irritation.
EUH208	Contains: FATTY ACIDS, TALLOIL, COMPOSED WITH OLEYLAMINE May produce an allergic reaction.

Precautionary statements:

P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P331	Do NOT induce vomiting.
P280	Wear protective gloves/ protective clothing / eye protection / face protection.
P301+P310	IF SWALLOWED: Immediately call a POISON CENTER or doctor.

Contains: XYLENE
MASS REACTION BETWEEN ETHYLBENZENE AND XYLENE

Product not intended for uses provided for by Directive 2004/42/EC.

2.3. Other hazards

On the basis of available data, the product does not contain any PBT or vPvB in percentage ≥ than 0,1%.

The product does not contain substances with endocrine disrupting properties in concentration ≥ 0.1%.

SECTION 3. Composition/information on ingredients

3.2. Mixtures

Contains:

Identification		x = Conc. %	Classification (EC) 1272/2008 (CLP)
N-BUTYL ACETATE			
INDEX	*607-025-00-1*	14 ≤ x < 19	**Flam. Liq. 3 H226, STOT SE 3 H336, EUH066**
EC	*204-658-1*		
CAS	*123-86-4*		
REACH Reg.	*01-2119485493-XXXX*		
XYLENE			
INDEX	*601-022-00-9*	10 ≤ x < 15	**Flam. Liq. 3 H226, Acute Tox. 4 H312, Acute Tox. 4 H332, Asp. Tox. 1 H304, STOT RE 2 H373, Skin Irrit. 2 H315, STOT SE 3 H335, Classification note according to Annex VI to the CLP Regulation: C**
EC	*215-535-7*		**ATE Dermal: 1100 mg/kg, ATE Inhalation vapours: 11 mg/l**
CAS	*1330-20-7*		
REACH Reg.	*01-2119488216-XXXX*		
BARIUM SULFATE			
INDEX		7 ≤ x < 9	
EC	*231-784-4*		
CAS	*7727-43-7*		
REACH Reg.	*01-2119491274-XXXX*		
2-METHOXY-1-METHYLETHYL ACETATE			
INDEX	*607-195-00-7*	7 ≤ x < 9	**Flam. Liq. 3 H226**
EC	*203-603-9*		
CAS	*108-65-6*		
REACH Reg.	*01-2119475791-XXXX*		
TITANIUM DIOXIDE			
INDEX		4,5 ≤ x < 5	**Classification note according to Annex VI to the CLP Regulation: 10**
EC	*236-675-5*		
CAS	*13463-67-7*		
REACH Reg.	*01-2119489379-XXXX*		
MASS REACTION BETWEEN ETHYLBENZENE AND XYLENE			
INDEX		2 ≤ x < 2,5	**Flam. Liq. 3 H226, Acute Tox. 4 H312, Acute Tox. 4 H332, Asp. Tox. 1 H304, STOT RE 2 H373, Eye Irrit. 2 H319, Skin Irrit. 2 H315, STOT SE 3 H335**
EC	*905-588-0*		**ATE Dermal: 1100 mg/kg, ATE Inhalation vapours: 11 mg/l**
CAS			

SECTION 3. Composition/information on ingredients ... / >>

REACH Reg. *01-2119488216-XXXX*
CARBON BLACK
INDEX 1 ≤ x < 1,5
EC *215-609-9*
CAS *1333-86-4*
REACH Reg. *01-2119384822-XXXX*
AMORPHOUS SILICATE HYDRATE
INDEX 0,35 ≤ x < 0,4
EC *231-545-4*
CAS *7631-86-9*
REACH Reg. *01-2119379499-XXXX*
PIGMENT YELLOW 110
INDEX 0,3 ≤ x < 0,35
EC *226-999-5*
CAS *5590-18-1*
ETHANOL
INDEX *603-002-00-5* 0,15 ≤ x < 0,2 **Flam. Liq. 2 H225**
EC *200-578-6*
CAS *64-17-5*
REACH Reg. *01-2119457610-XXXX*
FATTY ACIDS, C18-UNSATD., TRIMERS, COMPDS. WITH OLEYLAMINE
INDEX 0,05 ≤ x < 0,09 **Acute Tox. 4 H302, STOT RE 2 H373, Skin Sens. 1 H317, Aquatic Chronic 2 H411**
EC *604-612-4* **LD50 Oral: 1570 mg/kg**
CAS *147900-93-4*
REACH Reg. *01-2119971821-XXXX*
SOLVENT NAPHTHA (PETROLEUM), LIGHT AROMATIC
INDEX 0,05 ≤ x < 0,09 **Flam. Liq. 3 H226, Asp. Tox. 1 H304, STOT SE 3 H335, STOT SE 3 H336, Aquatic Chronic 2 H411, EUH066**
EC *918-668-5*
CAS *128601-23-0*
REACH Reg. *01-2119455851-XXXX*
FATTY ACIDS, TALLOIL, COMPOSED WITH OLEYLAMINE
INDEX 0,02 ≤ x < 0,06 **STOT RE 2 H373, Eye Dam. 1 H318, Skin Sens. 1A H317**
EC *288-315-1*
CAS *85711-55-3*
REACH Reg. *01-2119974148-XXXX*
ETHYLBENZENE
INDEX *601-023-00-4* 0 ≤ x < 0,03 **Flam. Liq. 2 H225, Acute Tox. 4 H332, Asp. Tox. 1 H304, STOT RE 2 H373, Aquatic Chronic 3 H412**
EC *202-849-4* **ATE Inhalation gas: 4500 ppm**
CAS *100-41-4*
REACH Reg. *01-2119489370-XXXX*
2,2-Dihydroxymethyl butanol
INDEX 0 ≤ x < 0,03 **Repr. 2 H361fd**
EC *201-074-9*
CAS *77-99-6*
REACH Reg. *01-2119486799-XXXX*
BUTYL TIN DILAURATE
INDEX *050-030-00-3* 0 ≤ x < 0,03 **Muta. 2 H341, Repr. 1B H360FD, STOT RE 1 H372**
EC *201-039-8*
CAS *77-58-7*
REACH Reg. *01-2119557828-21-XXXX*
PIGMENT VIOLET 19
INDEX 0 ≤ x < 0,03
EC *213-879-2*
CAS *1047-16-1*
QUARTZ (SIO2)
INDEX 0 ≤ x < 0,03 **Substance with a community workplace exposure limit.**
EC *238-878-4*
CAS *14808-60-7*
PIGMENT RED 122
INDEX 0 ≤ x < 0,03
EC *213-561-3*
CAS *980-26-7*

SECTION 3. Composition/information on ingredients ... / >>

4-MORPHOLINECARBALDEHYDE

INDEX		0 ≤ x < 0,03	**Skin Sens. 1 H317**
EC	*224-518-3*		
CAS	*4394-85-8*		
REACH Reg.	*01-2119987993-XXXX*		

LITHIUM CHLORIDE

INDEX		0 ≤ x < 0,03	**Acute Tox. 4 H302, Eye Irrit. 2 H319, Skin Irrit. 2 H315**
EC	*231-212-3*		**LD50 Oral: 526 mg/kg**
CAS	*7447-41-8*		
REACH Reg.	*01-2119560574-XXXX*		

The full wording of hazard (H) phrases is given in section 16 of the sheet.

SECTION 4. First aid measures

4.1. Description of first aid measures

EYES: Remove contact lenses, if present. Wash immediately with plenty of water for at least 15 minutes, opening the eyelids fully. If problem persists, seek medical advice.
SKIN: Remove contaminated clothing. Rinse skin with a shower immediately. Get medical advice/attention immediately. Wash contaminated clothing before using it again.
INHALATION: Remove to open air. If the subject stops breathing, administer artificial respiration. Get medical advice/attention immediately.
INGESTION: Get medical advice/attention immediately. Do not induce vomiting. Do not administer anything not explicitly authorised by a doctor.

4.2. Most important symptoms and effects, both acute and delayed

Specific information on symptoms and effects caused by the product are unknown.

4.3. Indication of any immediate medical attention and special treatment needed

Information not available

SECTION 5. Firefighting measures

5.1. Extinguishing media

SUITABLE EXTINGUISHING EQUIPMENT
The extinguishing equipment should be of the conventional kind: carbon dioxide, foam, powder and water spray.
UNSUITABLE EXTINGUISHING EQUIPMENT
None in particular.

5.2. Special hazards arising from the substance or mixture

HAZARDS CAUSED BY EXPOSURE IN THE EVENT OF FIRE
Do not breathe combustion products.

5.3. Advice for firefighters

GENERAL INFORMATION
Use jets of water to cool the containers to prevent product decomposition and the development of substances potentially hazardous for health. Always wear full fire prevention gear. Collect extinguishing water to prevent it from draining into the sewer system. Dispose of contaminated water used for extinction and the remains of the fire according to applicable regulations.
SPECIAL PROTECTIVE EQUIPMENT FOR FIRE-FIGHTERS
Normal fire fighting clothing i.e. fire kit (BS EN 469), gloves (BS EN 659) and boots (HO specification A29 and A30) in combination with self-contained open circuit positive pressure compressed air breathing apparatus (BS EN 137).

SECTION 6. Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

Block the leakage if there is no hazard.
Wear suitable protective equipment (including personal protective equipment referred to under Section 8 of the safety data sheet) to prevent any contamination of skin, eyes and personal clothing. These indications apply for both processing staff and those involved in emergency procedures.

Send away individuals who are not suitably equipped. Use explosion-proof equipment. Eliminate all sources of ignition (cigarettes, flames,

SECTION 6. Accidental release measures ... / >>

sparks, etc.) from the leakage site.

6.2. Environmental precautions

The product must not penetrate into the sewer system or come into contact with surface water or ground water.

6.3. Methods and material for containment and cleaning up

Collect the leaked product into a suitable container. Evaluate the compatibility of the container to be used, by checking section 10. Absorb the remainder with inert absorbent material.
Make sure the leakage site is well aired. Contaminated material should be disposed of in compliance with the provisions set forth in point 13.

6.4. Reference to other sections

Any information on personal protection and disposal is given in sections 8 and 13.

SECTION 7. Handling and storage

7.1. Precautions for safe handling

Keep away from heat, sparks and naked flames; do not smoke or use matches or lighters. Without adequate ventilation, vapours may accumulate at ground level and, if ignited, catch fire even at a distance, with the danger of backfire. Avoid bunching of electrostatic charges. Do not eat, drink or smoke during use. Remove any contaminated clothes and personal protective equipment before entering places in which people eat. Avoid leakage of the product into the environment.

7.2. Conditions for safe storage, including any incompatibilities

Store only in the original container. Store in a cool and well ventilated place, keep far away from sources of heat, naked flames and sparks and other sources of ignition. Keep containers away from any incompatible materials, see section 10 for details.

2-METHOXY-1-METHYLETHYL ACETATE
Store in an inert atmosphere, sheletered from moisture because it hydrolises easily.

Storage class TRGS 510 (Germany): 3

7.3. Specific end use(s)

Information not available

SECTION 8. Exposure controls/personal protection

8.1. Control parameters

Regulatory references:

BGR	България	НАРЕДБА № 13 ОТ 30 ДЕКЕМВРИ 2003 Г. ЗА ЗАЩИТА НА РАБОТЕЩИТЕ ОТ РИСКОВЕ, СВЪРЗАНИ С ЕКСПОЗИЦИЯ НА ХИМИЧНИ АГЕНТИ ПРИ РАБОТА (изм. ДВ. бр.5 от 17 Януари 2020г.)
CZE	Česká Republika	NAŘÍZENÍ VLÁDY ze dne 10. května 2021, kterým se mění nařízení vlády č. 361/2007 Sb., kterým se stanoví podmínky ochrany zdraví při práci
DEU	Deutschland	Forschungsgemeinschaft MAK- und BAT-Werte-Liste 2022 Ständige Senatskommission zur Prüfung gesundheitsschädlicher Arbeitsstoffe Mitteilung 58
DNK	Danmark	Bekendtgørelse om grænseværdier for stoffer og materialer - BEK nr 1458 af 13/12/2019
ESP	España	Límites de exposición profesional para agentes químicos en España 2023
EST	Eesti	Ohtlike kemikaalide ja neid sisaldavate materjalide kasutamise töötervishoiu ja tööohutuse nõuded ning töökeskkonna keemiliste ohutegurite piirnormid [RT I, 21.12.2022, 14]
FRA	France	Valeurs limites d'exposition professionnelle aux agents chimiques en FranceDécret n° 2021-1849 du 28 décembre 2021
GRC	Ελλάδα	Π.Δ. 26/2020 (ΦΕΚ 50/Α` 6.3.2020) Εναρμόνιση της ελληνικής νομοθεσίας προς τις διατάξεις των οδηγιών 2017/2398/ΕΕ, 2019/130/ΕΕ και 2019/983/ΕΕ «για την τροποποίηση της οδηγίας 2004/37/ΕΚ ''σχετικά με την προστασία των εργαζομένων από τους κινδύνους που συνδέονται με την έκθεση σε καρκινογόνους ή μεταλλαξιγόνους παράγοντες κατά την εργασία''»
HUN	Magyarország	Az innovációért és technológiáért felelős miniszter 5/2020. (II. 6.) ITM rendelete a kémiai kóroki tényezők hatásának kitett munkavállalók egészségének és biztonságának védelméről
HRV	Hrvatska	Pravilnik o izmjenama i dopunama Pravilnika o zaštiti radnika od izloženosti opasnimkemikalijama na radu, graničnim vrijednostima izloženosti i biološkim graničnim vrijednostima (NN 1/2021)
ITA	Italia	Decreto Legislativo 9 Aprile 2008, n.81
LVA	Latvija	Grozījumi Ministru kabineta 2007. gada 15. maija noteikumos Nr. 325 "Darba aizsardzības

SECTION 8. Exposure controls/personal protection ... / >>

		prasības saskarē ar ķīmiskajām vielām darba vietās" (prot. Nr. 32 18. §; prot. Nr. 1 22. §)
NOR	Norge	Forskrift om endring i forskrift om tiltaksverdier og grenseverdier for fysiske og kjemiske faktorer i arbeidsmiljøet samt smitterisikogrupper for biologiske faktorer (forskrift om tiltaks- og grenseverdier), 21. august 2018 nr. 1255
NLD	Nederland	Arbeidsomstandighedenregeling. Lijst van wettelijke grenswaarden op grond van de artikelen 4.3, eerste lid, en 4.16, eerste lid, van het Arbeidsomstandighedenbesluit
PRT	Portugal	Decreto-Lei n.º 1/2021 de 6 de janeiro, valores-limite de exposição profissional indicativos para os agentes químicos. Decreto-Lei n.º 35/2020 de 13 de julho, proteção dos trabalhadores contra os riscos ligados à exposição durante o trabalho a agentes cancerígenos ou mutagénicos
POL	Polska	Rozporządzenie ministra rozwoju, pracy i technologii z dnia 18 lutego 2021 r. Zmieniające rozporządzenie w sprawie najwyższych dopuszczalnych stężeń i natężeń czynników szkodliwych dla zdrowia w środowisku pracy
ROU	România	Hotărârea nr. 53/2021 pentru modificarea hotărârii guvernului nr. 1.218/2006, precum și pentru modificarea și completarea hotărârii guvernului nr. 1.093/2006
SWE	Sverige	Hygieniska gränsvärden, Arbetsmiljöverkets föreskrifter och allmänna råd om hygieniska gränsvärden (AFS 2018:1)
SVK	Slovensko	NARIADENIE VLÁDY Slovenskej republiky z 12. augusta 2020, ktorým sa mení a dopĺňa nariadenie vlády Slovenskej republiky č. 356/2006 Z. z. o ochrane zdravia zamestnancov pred rizikami súvisiacimi s expozíciou karcinogénnym a mutagénnym faktorom pri práci v znení neskorších predpisov
SVN	Slovenija	Pravilnik o varovanju delavcev pred tveganji zaradi izpostavljenosti kemičnim snovem pri delu (Uradni list RS, št. 100/01, 39/05, 53/07, 102/10, 43/11 – ZVZD-1, 38/15, 78/18 in 78/19)
TUR	Türkiye	Kimyasal Maddelerle Çalışmalarda Sağlık ve Güvenlik Önlemleri Hakkında Yönetmelik 12.08.2013 / 28733
GBR	United Kingdom	EH40/2005 Workplace exposure limits (Fourth Edition 2020)
EU	OEL EU	Directive (EU) 2022/431; Directive (EU) 2019/1831; Directive (EU) 2019/130; Directive (EU) 2019/983; Directive (EU) 2017/2398; Directive (EU) 2017/164; Directive 2009/161/EU; Directive 2006/15/EC; Directive 2004/37/EC; Directive 2000/39/EC; Directive 98/24/EC; Directive 91/322/EEC.
	TLV-ACGIH	ACGIH 2023

PIGMENT VIOLET 19

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral				25 mg/kg				
Inhalation						3 mg/m3		147 mg/m3
Skin				25 mg/kg				42 mg/kg

SECTION 8. Exposure controls/personal protection ... / >>

CARBON BLACK

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations
		mg/m3	ppm	mg/m3	ppm	
VLA	ESP	3,5				
VLEP	FRA	3,5				
VLEP	ITA	3				
WEL	GBR	3,5				
TLV-ACGIH		3				

Predicted no-effect concentration - PNEC

Normal value in fresh water	50	mg/l
Normal value in marine water		
Normal value for fresh water sediment	NPI	
Normal value for marine water sediment	NPI	
Normal value for water, intermittent release		
Normal value of STP microorganisms	VND	
Normal value for the food chain (secondary poisoning)	NPI	
Normal value for the terrestrial compartment	NPI	
Normal value for the atmosphere	NPI	

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Inhalation				0,060 mg/m3				1 mg/m3

FATTY ACIDS, C18-UNSATD., TRIMERS, COMPDS. WITH OLEYLAMINE

Predicted no-effect concentration - PNEC

Normal value in fresh water	0,006	mg/l
Normal value in marine water	0,0006	mg/l
Normal value for fresh water sediment	2,46	mg/kg
Normal value for marine water sediment	0,25	mg/kg
Normal value of STP microorganisms	NPI	
Normal value for the food chain (secondary poisoning)	0,47	mg/kg
Normal value for the terrestrial compartment	0,28	mg/kg
Normal value for the atmosphere	NPI	

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		NPI		0,012 mg/kg bw/d				
Skin	HIGH	NPI	HIGH	0,012 mg/kg bw/d	HIGH	NPI	HIGH	0,024 mg/kg bw/d

QUARTZ (SIO2)

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations
		mg/m3	ppm	mg/m3	ppm	
OEL	EU	0,1				
TLV-ACGIH		0,025				

SECTION 8. Exposure controls/personal protection ... / >>

4-MORPHOLINECARBALDEHYDE

Predicted no-effect concentration - PNEC

Normal value in fresh water	0,5	mg/l
Normal value in marine water	0,05	mg/l
Normal value for fresh water sediment	1,85	mg/kg
Normal value for marine water sediment	0,185	mg/kg
Normal value for marine water, intermittent release	5	mg/l
Normal value of STP microorganisms	2000	mg/l
Normal value for the terrestrial compartment	0,0764	mg/kg
Normal value for the atmosphere	NPI	

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		NPI		4,17 mg/kg bw/d				
Inhalation	NPI	NPI	13,3 mg/m3	8,93 mg/m3	NPI	NPI	13,3 mg/m3	50,3 mg/m3
Skin	NPI	NPI	MED	4,17 mg/kg bw/d	NPI	NPI	MED	11,7 mg/kg bw/d

PIGMENT YELLOW 110

Predicted no-effect concentration - PNEC

Normal value in fresh water	NPI	
Normal value in marine water	NPI	
Normal value for fresh water sediment	NPI	
Normal value for marine water sediment	NPI	
Normal value for water, intermittent release	NPI	
Normal value of STP microorganisms	NPI	
Normal value for the terrestrial compartment	NPI	
Normal value for the atmosphere	NPI	

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		NPI		NPI				
Inhalation	NPI	NPI	NPI	NPI	NPI	NPI	NPI	NPI
Skin	NPI	NPI	NPI	NPI	NPI	NPI	NPI	NPI

LITHIUM CHLORIDE

Predicted no-effect concentration - PNEC

Normal value in fresh water	10,4	mg/l
Normal value in marine water	1,04	mg/l
Normal value for fresh water sediment	49,9	mg/kg
Normal value for marine water sediment	4,99	mg/kg
Normal value for water, intermittent release	10,4	mg/l
Normal value of STP microorganisms	140,2	mg/l
Normal value for the terrestrial compartment	4,13	mg/kg

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		21,96 mg/kg/d		7,32 mg/kg/d				
Inhalation		30 mg/m3		10 mg/m3		30 mg/m3		10 mg/m3
Skin		50 mg/kg/d		72,3 mg/kg/d		100 mg/kg/d		73,2 mg/kg/d

SECTION 8. Exposure controls/personal protection ... / >>

BUTYL TIN DILAURATE

Predicted no-effect concentration - PNEC

Normal value in fresh water	0,00046	mg/l
Normal value in marine water	0,000043	mg/l
Normal value for fresh water sediment	0,0463	mg/kg
Normal value for marine water sediment	0,005	mg/kg
Normal value for marine water, intermittent release	0,00463	mg/l
Normal value for fresh water, intermittent release	0,00463	mg/l
Normal value of STP microorganisms	100	mg/l
Normal value for the food chain (secondary poisoning)	0,2	mg/kg
Normal value for the terrestrial compartment	0,0407	mg/kg

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		0.020 mg/kg/bw		0.0031 mg/kg/bw				
Inhalation	NPI	0.040 mg/m3	NPI	0.0046 mg/m3	NPI	0.059 mg/m3	NPI	0,02 mg/m3
Skin	NPI	0.50 mg/kg/bw	NPI	0.160 mg/kg/bw	NPI	2.08 mg/kg	NPI	0.430 mg/kg/bw

2,2-Dihydroxymethyl butanol

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations
		mg/m3	ppm	mg/m3	ppm	
NGV/KGV	SWE	5				

Predicted no-effect concentration - PNEC

Normal value in fresh water	NPI
Normal value in marine water	NPI
Normal value for fresh water sediment	NPI
Normal value for marine water sediment	NPI
Normal value for water, intermittent release	NPI
Normal value of STP microorganisms	NPI
Normal value for the food chain (secondary poisoning)	NPI
Normal value for the terrestrial compartment	NPI
Normal value for the atmosphere	NPI

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		NPI		0,340 mg/kg bw/d				
Inhalation	NPI	NPI	NPI	0,580 mg/m3	NPI	NPI	NPI	3,3 mg/m3
Skin	NPI	NPI	NPI	0,340 mg/kg bw/d	NPI	NPI	NPI	0,940 mg/kg bw/d

SECTION 8. Exposure controls/personal protection ... / >>

FATTY ACIDS, TALLOIL, COMPOSED WITH OLEYLAMINE

Predicted no-effect concentration - PNEC

Normal value in fresh water	NPI	
Normal value in marine water	NPI	
Normal value for fresh water sediment	NEA	
Normal value for marine water sediment	NEA	
Normal value for water, intermittent release	NEA	
Normal value of STP microorganisms	NPI	
Normal value for the food chain (secondary poisoning)	0,47	mg/kg
Normal value for the terrestrial compartment	NEA	
Normal value for the atmosphere	NPI	

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		NPI		0,012 mg/kg bw/d				
Skin	HIGH	NPI	HIGH	0,012 mg/kg bw/d	HIGH	NPI	HIGH	0,024 mg/kg bw/d

MASS REACTION BETWEEN ETHYLBENZENE AND XYLENE

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations
		mg/m3	ppm	mg/m3	ppm	
OEL	EU	221	50	442	100	
TLV-ACGIH			100		150	A4,BEI-URT and eye irr, CNS

Predicted no-effect concentration - PNEC

Normal value in fresh water	0,327	mg/l
Normal value in marine water	0,327	mg/l
Normal value for fresh water sediment	12,46	mg/kg
Normal value for marine water sediment	12,46	mg/kg
Normal value for water, intermittent release	0,327	mg/l
Normal value of STP microorganisms	6,58	mg/l
Normal value for the terrestrial compartment	2,31	mg/kg

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		NPI		12,5 mg/kg bw/d				
Inhalation	260 mg/m3	260 mg/m3	65,3 mg/m3	65,3 mg/m3	442 mg/m3	442 mg/m3	221 mg/m3	221 mg/m3
Skin	LOW	LOW	NPI	125 mg/kg bw/d	LOW	LOW	NPI	212 mg/kg bw/d

PIGMENT RED 122

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral				25 mg/kg				
Inhalation							3 mg/m3	147 mg/m3
Skin				25 mg/kg				42 mg/kg

SECTION 8. Exposure controls/personal protection ... / >>

SOLVENT NAPHTHA (PETROLEUM), LIGHT AROMATIC

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations
		mg/m3	ppm	mg/m3	ppm	
TLV	DNK		10			provisional
TLV-ACGIH		100	19			

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral				7,5 mg/kg bw/d				
Inhalation				32 mg/m3				151 mg/m3
Skin				7,5 mg/kg bw/d				12,5 mg/kg bw/d

XYLENE

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations	
		mg/m3	ppm	mg/m3	ppm		
TLV	BGR	221	50	442	100	SKIN	
TLV	CZE	200	45,4	400	90,8	SKIN	
AGW	DEU	220	50	440	100	SKIN	
MAK	DEU	220	50	440	100	SKIN	
TLV	DNK	109	25			SKIN	E
VLA	ESP	221	50	442	100	SKIN	
TLV	EST	200	50	450	100	SKIN	
VLEP	FRA	221	50	442	100	SKIN	
TLV	GRC	435	100	650	150		
AK	HUN	221	50	442	100	SKIN	
GVI/KGVI	HRV	221	50	442	100	SKIN	
VLEP	ITA	221	50	442	100	SKIN	
RV	LVA	221	50	442	100	SKIN	
TLV	NOR	108	25			SKIN	
TGG	NLD	210		442		SKIN	
VLE	PRT	221	50	442	100	SKIN	
NDS/NDSCh	POL	100		200		SKIN	
TLV	ROU	221	50	442	100	SKIN	
NGV/KGV	SWE	221	50	442	100	SKIN	
NPEL	SVK	221	50	442	100	SKIN	
MV	SVN	221	50	442	100	SKIN	
ESD	TUR	221	50	442	100	SKIN	
WEL	GBR	220	50	441	100	SKIN	
OEL	EU	221	50	442	100	SKIN	
TLV-ACGIH			20				

Predicted no-effect concentration - PNEC

Normal value in fresh water	0,327	mg/l
Normal value in marine water	0,327	mg/l
Normal value for fresh water sediment	12,46	mg/kg
Normal value for marine water sediment	12,46	mg/kg
Normal value for water, intermittent release	0,327	mg/l
Normal value of STP microorganisms	6,58	mg/l
Normal value for the terrestrial compartment	2,31	mg/kg

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral				1,5 mg/kg/bw				
Inhalation	260 mg/m3	260 mg/m3		65,3 mg/m3	442 mg/m3	442 mg/m3	77 mg/m3	221 mg/m3
Skin				125 mg/kg/bw	174 mg/m3	180 mg/kg bw/d		212 mg/kg bw/d

SECTION 8. Exposure controls/personal protection ... / >>

AMORPHOUS SILICATE HYDRATE

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations
		mg/m3	ppm	mg/m3	ppm	
AGW	DEU	4				INHAL
MAK	DEU	4				INHAL
MAK	DEU	0,02		0,16		RESP
TLV	EST	2				
RV	LVA	1				
MV	SVN	4				INHAL

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Inhalation								4 mg/m3

TITANIUM DIOXIDE

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations	
		mg/m3	ppm	mg/m3	ppm		
TLV	BGR	10				RESP	
MAK	DEU	0,3		2,4		RESP	Hinweis
TLV	DNK	6					Som Ti
VLA	ESP	10					
TLV	EST	5					
VLEP	FRA	10					
TLV	GRC		10				
GVI/KGVI	HRV	10				INHAL	
GVI/KGVI	HRV	4				RESP	
RV	LVA	10					
TLV	NOR	5					
NDS/NDSCh	POL	10				INHAL	
TLV	ROU	10		15			
NGV/KGV	SWE	5					Totaldamm
NPEL	SVK	5					
WEL	GBR	10				INHAL	
WEL	GBR	4				RESP	
TLV-ACGIH		0,2				RESP	

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		NPI		NPI				
Inhalation	NPI	NPI	0,028 mg/m3	NPI	NPI	NPI	0,170 mg/m3	NPI
Skin	NPI	NPI	NPI	NPI	NPI	NPI	NPI	NPI

SECTION 8. Exposure controls/personal protection ... / >>

BARIUM SULFATE

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations	
		mg/m3	ppm	mg/m3	ppm		
TLV	BGR	10					
MAK	DEU	4				INHAL	
MAK	DEU	0,3		2,4		RESP	Hinweis
VLA	ESP	10					
GVI/KGVI	HRV	10				INHAL	
GVI/KGVI	HRV	4				RESP	
NDS/NDSCh	POL	0,5					Na Ba
NPEL	SVK	4				INHAL	
NPEL	SVK	1,5				RESP	
WEL	GBR	10				INHAL	
WEL	GBR	4				RESP	
TLV-ACGIH		5				INHAL	

Predicted no-effect concentration - PNEC

Normal value in fresh water	0,115	mg/l
Normal value for fresh water sediment	600,4	mg/kg
Normal value of STP microorganisms	62,2	mg/l
Normal value for the terrestrial compartment	207,7	mg/kg

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral				13000 mg/kg				
Inhalation				10 mg/m3			10 mg/m3	10 mg/m3

SECTION 8. Exposure controls/personal protection ... / >>

2-METHOXY-1-METHYLETHYL ACETATE

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations
		mg/m3	ppm	mg/m3	ppm	
TLV	BGR	275	50	550	100	SKIN
TLV	CZE	270	49,14	550	100,1	SKIN
AGW	DEU	270	50	270	50	
MAK	DEU	270	50	270	50	
TLV	DNK	275	50	550	100	SKIN E
VLA	ESP	275	50	550	100	SKIN
TLV	EST	275	50	550	100	SKIN
VLEP	FRA	275	50	550	100	SKIN
TLV	GRC	275	50	550	100	
AK	HUN	275	50	550	100	
GVI/KGVI	HRV	275	50	550	100	SKIN
VLEP	ITA	275	50	550	100	SKIN
RV	LVA	275	50	550	100	SKIN
TLV	NOR	270	50			SKIN
TGG	NLD	550				
VLE	PRT	275	50	550	100	SKIN
NDS/NDSCh	POL	260		520		SKIN
TLV	ROU	275	50	550	100	SKIN
NGV/KGV	SWE	275	50	550	100	SKIN
NPEL	SVK	275	50	550	100	SKIN
MV	SVN	275	50	550	100	SKIN
ESD	TUR	275	50	550	100	SKIN
WEL	GBR	274	50	548	100	SKIN
OEL	EU	275	50	550	100	SKIN

Predicted no-effect concentration - PNEC

Normal value in fresh water	0,635	mg/l
Normal value in marine water	0,0635	mg/l
Normal value for fresh water sediment	3,29	mg/kg
Normal value for marine water sediment	0,329	mg/kg
Normal value for marine water, intermittent release	6,35	mg/l
Normal value of STP microorganisms	100	mg/l
Normal value for the terrestrial compartment	0,29	mg/kg
Normal value for the atmosphere	NPI	

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		500 mg/kg bw/d		36 mg/kg				
Inhalation	NPI	NPI	33 mg/m3	33 mg/m3	550 mg/m3	NPI	NPI	275 mg/m3
Skin	NPI	NPI	NPI	320 mg/kg	NPI	NPI	NPI	796 mg/kg

SECTION 8. Exposure controls/personal protection ... / >>

ETHYLBENZENE

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations
		mg/m3	ppm	mg/m3	ppm	
TLV	BGR	435		545		SKIN
TLV	CZE	200	45,4	500	113,5	SKIN
AGW	DEU	88	20	176	40	SKIN
MAK	DEU	88	20	176	40	SKIN
TLV	DNK	217	50	434	100	SKIN E
VLA	ESP	441	100	884	200	SKIN
TLV	EST	442	100	884	200	SKIN
VLEP	FRA	88,4	20	442	100	SKIN
TLV	GRC	435	100	545	125	
AK	HUN	442	100	884	200	SKIN
GVI/KGVI	HRV	442	100	884	200	SKIN
VLEP	ITA	442	100	884	200	SKIN
RV	LVA	442	100	884	200	SKIN
TLV	NOR	20	5			SKIN
TGG	NLD	215		430		SKIN
VLE	PRT	442	100	884	200	SKIN
NDS/NDSCh	POL	200		400		SKIN
TLV	ROU	442	100	884	200	SKIN
NGV/KGV	SWE	220	50	884	200	SKIN
NPEL	SVK	442	100	884	200	SKIN
MV	SVN	442	100	884	200	SKIN
ESD	TUR	442	100	884	200	SKIN
WEL	GBR	441	100	552	125	SKIN
OEL	EU	442	100	884	200	SKIN
TLV-ACGIH		87	20			

Predicted no-effect concentration - PNEC

Normal value in fresh water	0,1	mg/l
Normal value in marine water	0,01	mg/l
Normal value for fresh water sediment	13,7	mg/kg
Normal value for marine water sediment	1,37	mg/kg
Normal value for marine water, intermittent release	0,1	mg/l
Normal value of STP microorganisms	9,6	mg/l
Normal value for the food chain (secondary poisoning)	20	mg/kg
Normal value for the terrestrial compartment	2,68	mg/kg
Normal value for the atmosphere	NPI	

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		NPI		1,6 mg/kg bw/d				
Inhalation	NPI	LOW	NPI	15 mg/m3	293 mg/m3	LOW	442 mg/m3	77 mg/m3
Skin	NPI	NPI	NPI	NPI	NPI	NPI	NPI	180 mg/kg bw/d

SECTION 8. Exposure controls/personal protection ... / >>

ETHANOL

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations
		mg/m3	ppm	mg/m3	ppm	
TLV	BGR	1000				
TLV	CZE	1000	522	3000	1566	
AGW	DEU	380	200	1520	800	
MAK	DEU	380	200	1520	800	
TLV	DNK	1900	1000			
VLA	ESP			1910	1000	
TLV	EST	1000	500	1900	1000	
VLEP	FRA	1900	1000	9500	5000	
TLV	GRC	1900	1000			
AK	HUN	1900	1000	3800	2000	
GVI/KGVI	HRV	1900	1000			
RV	LVA	1000				
TLV	NOR	950	500			
TGG	NLD	260		1900		SKIN
NDS/NDSCh	POL	1900				
TLV	ROU	1900	1000	9500	5000	
NGV/KGV	SWE	1000	500	1900 (C)	1000 (C)	
NPEL	SVK	960	500	1920	1000	
MV	SVN	960	500	1920	1000	
WEL	GBR	1920	1000			
TLV-ACGIH				1884	1000	

Predicted no-effect concentration - PNEC

Normal value in fresh water	0,96	mg/l
Normal value in marine water	0,79	mg/l
Normal value for fresh water sediment	3,6	mg/kg
Normal value for marine water sediment	2,9	mg/l
Normal value for water, intermittent release	2,75	mg/l
Normal value of STP microorganisms	580	mg/l
Normal value for the food chain (secondary poisoning)	0,38	mg/kg
Normal value for the terrestrial compartment	0,63	mg/kg
Normal value for the atmosphere	NPI	

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		NPI		87 mg/kg				
Inhalation	950 mg/m3	NPI	NPI	114 mg/m3	1900 mg/m3	NPI	NPI	380 mg/m3
Skin	NPI	NPI	NPI	206 mg/kg	NPI	NPI	NPI	343 mg/kg

SECTION 8. Exposure controls/personal protection ... / >>

N-BUTYL ACETATE

Threshold Limit Value

Type	Country	TWA/8h		STEL/15min		Remarks / Observations
		mg/m3	ppm	mg/m3	ppm	
TLV	BGR	710		950		
TLV	CZE	241		723		
AGW	DEU	300	62	600	124	
MAK	DEU	480	100	960	200	
TLV	DNK	241	50	723	150	E
VLA	ESP	241	50	723	150	
TLV	EST	500	100	700	150	
VLEP	FRA	241	50	723	150	
TLV	GRC	710	150	950	200	
AK	HUN	241	50	723	150	
GVI/KGVI	HRV	241	50	723	150	
VLEP	ITA	241	50	723	150	
RV	LVA	200				
TLV	NOR		75			
TGG	NLD	150				
VLE	PRT	241	50	723	150	
NDS/NDSCh	POL	240		720		
TLV	ROU	241	50	723	150	
NGV/KGV	SWE	241	50	723 (C)	150 (C)	
NPEL	SVK	241	50	723	150	
MV	SVN	300	62	600	124	
WEL	GBR	724	150	966	200	
OEL	EU	241	50	723	150	
TLV-ACGIH			50		150	

Predicted no-effect concentration - PNEC

Normal value in fresh water	0,18	mg/l
Normal value in marine water	0,018	mg/l
Normal value for fresh water sediment	0,981	mg/kg
Normal value for marine water sediment	0,0981	mg/kg
Normal value for marine water, intermittent release	0,36	mg/l
Normal value of STP microorganisms	35,6	mg/l
Normal value for the terrestrial compartment	0,0903	mg/kg
Normal value for the atmosphere	NPI	

Health - Derived no-effect level - DNEL / DMEL

	Effects on consumers				Effects on workers			
Route of exposure	Acute local	Acute systemic	Chronic local	Chronic systemic	Acute local	Acute systemic	Chronic local	Chronic systemic
Oral		2 mg/kg/d		2 mg/kg/d				
Inhalation	300 mg/m3	300 mg/m3	35,7 mg/m3	35,7 mg/m3	600 mg/m3	600 mg/m3	300 mg/m3	300 mg/m3
Skin	NPI	6 mg/kg/d	NPI	6 mg/kg/d	NPI	11 mg/kg/d	NPI	11 mg/kg/d

Legend:
(C) = CEILING ; INHAL = Inhalable Fraction ; RESP = Respirable Fraction ; THORA = Thoracic Fraction.
VND = hazard identified but no DNEL/PNEC available ; NEA = no exposure expected ; NPI = no hazard identified ; LOW = low hazard ; MED = medium hazard ; HIGH = high hazard.

8.2. Exposure controls

As the use of adequate technical equipment must always take priority over personal protective equipment, make sure that the workplace is well aired through effective local aspiration.
When choosing personal protective equipment, ask your chemical substance supplier for advice.
Personal protective equipment must be CE marked, showing that it complies with applicable standards.
Provide an emergency shower with face and eye wash station.
Exposure levels must be kept as low as possible to avoid significant build-up in the organism. Manage personal protective equipment so as to guarantee maximum protection (e.g. reduction in replacement times).
HAND PROTECTION
Protect hands with category III work gloves.
The following should be considered when choosing work glove material (see standard EN 374): compatibility, degradation, permeability time.
The work gloves' resistance to chemical agents should be checked before use, as it can be unpredictable. The gloves' wear time depends on the duration and type of use.
SKIN PROTECTION

SECTION 8. Exposure controls/personal protection ... / >>

Wear category II professional long-sleeved overalls and safety footwear (see Regulation 2016/425 and standard EN ISO 20344). Wash body with soap and water after removing protective clothing.
Consider the appropriateness of providing antistatic clothing in the case of working environments in which there is a risk of explosion.
EYE PROTECTION
Wear airtight protective goggles (see standard EN ISO 16321).
RESPIRATORY PROTECTION
Respiratory protection devices must be used if the technical measures adopted are not suitable for restricting the worker's exposure to the threshold values considered. Use a mask with a type A filter whose class (1, 2 or 3) must be chosen according to the limit of use concentration. (see standard EN 14387).
If the substance considered is odourless or its olfactory threshold is higher than the corresponding TLV-TWA and in the case of an emergency, wear open-circuit compressed air breathing apparatus (in compliance with standard EN 137) or external air-intake breathing apparatus (in compliance with standard EN 138). For a correct choice of respiratory protection device, see standard EN 529.
ENVIRONMENTAL EXPOSURE CONTROLS
The emissions generated by manufacturing processes, including those generated by ventilation equipment, should be checked to ensure compliance with environmental standards.

SECTION 9. Physical and chemical properties

9.1. Information on basic physical and chemical properties

Properties	Value	Information
Appearance	liquid	
Colour	grey	
Odour	solvent	
Melting point / freezing point	-77 °C	
Initial boiling point	126 °C	
Flammability	flammable liquid	
Lower explosive limit	1 % (v/v)	Temperature: 20 °C
Upper explosive limit	7,6 % (v/v)	Temperature: 20 °C
Flash point	17 °C	
Auto-ignition temperature	464 °C	
Decomposition temperature	not applicable	
pH	not applicable	
Kinematic viscosity	not applicable	
Solubility	soluble in organic solvents	
Partition coefficient: n-octanol/water	not applicable	
Vapour pressure	5,98 kPa	Temperature: 20 °C
Density and/or relative density	1,05 g/cm3	Temperature: 20 °C
Relative vapour density	3,66	
Particle characteristics	not applicable	

9.2. Other information

9.2.1. Information with regard to physical hazard classes

Information not available

9.2.2. Other safety characteristics

VOC (Directive 2010/75/EU)	36,28 %	-	380,90	g/litre

SECTION 10. Stability and reactivity

10.1. Reactivity

There are no particular risks of reaction with other substances in normal conditions of use.

2-METHOXY-1-METHYLETHYL ACETATE
Stable in normal conditions of use and storage.
With the air it may slowly develop peroxides that explode with an increase in temperature.

N-BUTYL ACETATE
Decomposes on contact with: water.

10.2. Chemical stability

SECTION 10. Stability and reactivity ... / >>

The product is stable in normal conditions of use and storage.

10.3. Possibility of hazardous reactions

The vapours may also form explosive mixtures with the air.

XYLENE
Stable in normal conditions of use and storage.Reacts violently with: strong oxidants,strong acids,nitric acid,perchlorates.May form explosive mixtures with: air.
2-METHOXY-1-METHYLETHYL ACETATE
May react violently with: oxidising substances,strong acids,alkaline metals.
ETHYLBENZENE
Reacts violently with: strong oxidants.Attacks various types of plastic materials.May form explosive mixtures with: air.
ETHANOL
Risk of explosion on contact with: alkaline metals,alkaline oxides,calcium hypochlorite,sulphur monofluoride,acetic anhydride,acids,concentrated hydrogen peroxide,perchlorates,perchloric acid,perchloronitrile,mercury nitrate,nitric acid,silver,silver nitrate,ammonia,silver oxide,ammonia,strong oxidising agents,nitrogen dioxide.May react dangerously with: bromoacetylene,chlorine acetylene,bromine trifluoride,chromium trioxide,chromyl chloride,fluorine,potassium tert-butoxide,lithium hydride,phosphorus trioxide,black platinum,zirconium (IV) chloride,zirconium (IV) iodide.Forms explosive mixtures with: air.
N-BUTYL ACETATE
Risk of explosion on contact with: strong oxidising agents.May react dangerously with: alkaline hydroxides,potassium tert-butoxide.Forms explosive mixtures with: air.

10.4. Conditions to avoid

Avoid overheating. Avoid bunching of electrostatic charges. Avoid all sources of ignition.

ETHANOL
Avoid exposure to: sources of heat,naked flames.
N-BUTYL ACETATE
Avoid exposure to: moisture,sources of heat,naked flames.

10.5. Incompatible materials

2-METHOXY-1-METHYLETHYL ACETATE
Incompatible with: oxidising substances,strong acids,alkaline metals.
N-BUTYL ACETATE
Incompatible with: water,nitrates,strong oxidants,acids,alkalis,zinc.

10.6. Hazardous decomposition products

In the event of thermal decomposition or fire, gases and vapours that are potentially dangerous to health may be released.

ETHYLBENZENE
May develop: methane,styrene,hydrogen,ethane.

SECTION 11. Toxicological information

In the absence of experimental data for the product itself, health hazards are evaluated according to the properties of the substances it contains, using the criteria specified in the applicable regulation for classification.
It is therefore necessary to take into account the concentration of the individual hazardous substances indicated in section 3, to evaluate the toxicological effects of exposure to the product.

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Metabolism, toxicokinetics, mechanism of action and other information

2-METHOXY-1-METHYLETHYL ACETATE
The main route of entry is the skin, whereas the respiratory route is less important due to the low vapour pressure of the product.

Information on likely routes of exposure

XYLENE
WORKERS: inhalation; contact with the skin.
POPULATION: ingestion of contaminated food or water; inhalation of ambient air.

2-METHOXY-1-METHYLETHYL ACETATE
WORKERS: inhalation; contact with the skin.

ETHYLBENZENE
WORKERS: inhalation; contact with the skin.
POPULATION: ingestion of contaminated food or water; contact with the skin of products containing the substance.

SECTION 11. Toxicological information ... / >>

N-BUTYL ACETATE
WORKERS: inhalation; contact with the skin.

Delayed and immediate effects as well as chronic effects from short and long-term exposure

XYLENE
Toxic effect on the central nervous system (encephalopathy); irritating for the skin, conjunctiva, cornea and respiratory apparatus.

2-METHOXY-1-METHYLETHYL ACETATE
Above 100 ppm causes irritation of the eye, nose and oropharynx mucous membranes. At 1000 ppm, disturbance of equilibrium and severe eye irritation can be noticed. Clinical and biological examinations carried out on exposed volunteers revealed no anomalies. Acetate produces greater skin and eye irritation with direct contact. No chronic effects on humans have been reported (INCR, 2010).

ETHYLBENZENE
As the counterparts of benzene, may have an acute effect on the central nervous system, with depression, narcosis, often preceded by dizziness and associated with headache (Ispesl). Is irritating for skin, conjunctiva and respiratory tract.

N-BUTYL ACETATE
In humans, the substance's vapours cause irritation of the eyes and nose. In the event of repeated exposure, skin irritation, dermatitis (dryness and cracking of the skin) and keratitis appear.

Interactive effects

XYLENE
Intake of alcohol interferes with the metabolism of the substance, inhibiting it. Ethanol consumption (0.8 g/kg) before a 4-hour exposure to xylene vapours (145 and 280 ppm) causes a 50% reduction in the excretion of methyl hippuric acid, whereas the concentration of xylenes in the blood increases approx. 1.5-2 times. At the same time there is an increase in the secondary side effects of the ethanol. The metabolism of the xylenes is increased by phenobarbital and 3-methyl-colantrene type enzyme inducers. Aspirin and xylenes mutually inhibit their conjugation with the glycine, which results in a decrease in urinary excretion of methyl hippuric acid. Other industrial products can interfere with the metabolism of xylenes.

N-BUTYL ACETATE
A case of acute intoxication been reported involving a 33 year old worker while cleaning a tank with a preparation containing xylenes, butyl acetate and ethylene glycol acetate. The person had irritation of the conjunctiva and upper respiratory tract, drowsiness and motor coordination disorders, which disappeared within 5 hours. The symptoms are attributed to poisoning by mixed xylenes and butyl acetate, with a possible synergistic effect responsible for the neurological effects. Cases of vacuolar keratitis are reported in workers exposed to a mixture of butyl acetate and isobutanol vapours, but with uncertainty concerning the responsibility of a particular solvent (INRC, 2011).

ACUTE TOXICITY

ATE (Inhalation - vapours) of the mixture:	> 20 mg/l
ATE (Oral) of the mixture:	Not classified (no significant component)
ATE (Dermal) of the mixture:	>2000 mg/kg

PIGMENT VIOLET 19

LD50 (Oral):	> 2000 mg/kg Rat

CARBON BLACK

LD50 (Oral):	> 8000 mg/kg Rat
LC50 (Inhalation vapours):	0,0046 mg/l/4h Rat

FATTY ACIDS, C18-UNSATD., TRIMERS, COMPDS. WITH OLEYLAMINE

LD50 (Oral):	1570 mg/kg Rat

4-MORPHOLINECARBALDEHYDE

LD50 (Dermal):	> 18400 mg/kg Rat
LD50 (Oral):	7314 mg/kg Rat
LC50 (Inhalation vapours):	> 5,319 mg/l/4h Rat

PIGMENT YELLOW 110

LD50 (Dermal):	> 5000 mg/kg Rat
LD50 (Oral):	> 2340 mg/kg Rat

LITHIUM CHLORIDE

LD50 (Dermal):	> 2000 mg/kg Rat
LD50 (Oral):	526 mg/kg Rat
LC50 (Inhalation vapours):	> 5,57 mg/l/4h Rat

SECTION 11. Toxicological information ... / >>

2,2-Dihydroxymethyl butanol
LD50 (Oral): > 14700 mg/kg Rat
LC50 (Inhalation mists/powders): > 0,85 mg/l/4h

FATTY ACIDS, TALLOIL, COMPOSED WITH OLEYLAMINE
LD50 (Oral): 2000 mg/kg Rat

MASS REACTION BETWEEN ETHYLBENZENE AND XYLENE
LD50 (Dermal): 12126 mg/kg rabbit
ATE (Dermal): 1100 mg/kg estimate from table 3.1.2 of Annex I of the CLP (figure used for calculation of the acute toxicity estimate of the mixture)
LD50 (Oral): > 3523 mg/kg rat
LC50 (Inhalation vapours): 6,35 mg/l/4h rat
ATE (Inhalation vapours): 11 mg/l estimate from table 3.1.2 of Annex I of the CLP (figure used for calculation of the acute toxicity estimate of the mixture)

PIGMENT RED 122
LD50 (Oral): > 2000 mg/kg Rat

SOLVENT NAPHTHA (PETROLEUM), LIGHT AROMATIC
LD50 (Dermal): > 2000 mg/kg Rabbit
LD50 (Oral): 3492 mg/kg Rat
LC50 (Inhalation vapours): > 6193 mg/l/4h Rat

UREA-ALDEHYDE RESIN
LD50 (Oral): > 5000 mg/kg Rat

XYLENE
LD50 (Dermal): > 4200 mg/kg Rabbit
ATE (Dermal): 1100 mg/kg estimate from table 3.1.2 of Annex I of the CLP (figure used for calculation of the acute toxicity estimate of the mixture)
LD50 (Oral): 4300 mg/kg Rat
LC50 (Inhalation vapours): 6700 ppm/4 Rat
ATE (Inhalation vapours): 11 mg/l estimate from table 3.1.2 of Annex I of the CLP (figure used for calculation of the acute toxicity estimate of the mixture)

AMORPHOUS SILICATE HYDRATE
LD50 (Dermal): > 2000 mg/kg Rat
LD50 (Oral): > 2000 mg/kg Rat
LC50 (Inhalation mists/powders): > 2,2 mg/l/1h Rat

TITANIUM DIOXIDE
LD50 (Oral): > 2000 mg/kg Rat
LC50 (Inhalation gas): > 3,43 mg/l/4h rat

BARIUM SULFATE
LD50 (Oral): > 3000 mg/kg Mouse

2-METHOXY-1-METHYLETHYL ACETATE
LD50 (Dermal): 2000 mg/kg Rat
LD50 (Oral): > 5155 mg/kg Rat

ETHYLBENZENE
LD50 (Dermal): 3500 mg/kg Rat
LD50 (Oral): 3500 mg/kg Rat
LC50 (Inhalation gas): 3500 mg/kg bw Rat

ETHANOL
LD50 (Oral): > 5000 mg/kg Rat
LC50 (Inhalation vapours): 117 mg/l/4h Rat

N-BUTYL ACETATE
LD50 (Dermal): > 14000 mg/kg bw/d Rabbit
LD50 (Oral): 10736 mg/kg Rat
LC50 (Inhalation vapours): > 21,1 mg/l/4h Rat

SKIN CORROSION / IRRITATION

Causes skin irritation

SECTION 11. Toxicological information ... / >>

SERIOUS EYE DAMAGE / IRRITATION

Does not meet the classification criteria for this hazard class

RESPIRATORY OR SKIN SENSITISATION

May produce an allergic reaction.
Contains:
FATTY ACIDS, TALLOIL, COMPOSED WITH OLEYLAMINE

GERM CELL MUTAGENICITY

Does not meet the classification criteria for this hazard class

CARCINOGENICITY

Does not meet the classification criteria for this hazard class

XYLENE
Classified in Group 3 (not classifiable as a human carcinogen) by the International Agency for Research on Cancer (IARC).
The US Environmental Protection Agency (EPA) affirms that "the data is inadequate for an assessment of the carcinogenic potential".

ETHYLBENZENE
Classified in Group 2B (possible human carcinogen) by the International Agency for Research on Cancer (IARC) - (IARC, 2000).
Classified in Group D (not classifiable as a human carcinogen) by the US Environmental Protection Agency (EPA) - (US EPA file on-line 2014).

REPRODUCTIVE TOXICITY

Does not meet the classification criteria for this hazard class

STOT - SINGLE EXPOSURE

Does not meet the classification criteria for this hazard class

STOT - REPEATED EXPOSURE

May cause damage to organs

ASPIRATION HAZARD

Toxic for aspiration

11.2. Information on other hazards

Based on the available data, the product does not contain substances listed in the main European lists of potential or suspected endocrine disruptors with human health effects under evaluation.

SECTION 12. Ecological information

Use this product according to good working practices. Avoid littering. Inform the competent authorities, should the product reach waterways or contaminate soil or vegetation.

12.1. Toxicity

CARBON BLACK

LC50 - for Fish	1000 mg/l/96h Brachydanio rerio
EC50 - for Crustacea	> 5600 mg/l/48h Daphnia magna
EC50 - for Algae / Aquatic Plants	> 10000 mg/l/72h Scenedesmus subspicatus
Chronic NOEC for Algae / Aquatic Plants	10000 mg/l Scenedesmus subspicatus

FATTY ACIDS, C18-UNSATD., TRIMERS, COMPDS. WITH OLEYLAMINE

EC50 - for Algae / Aquatic Plants	8 mg/l/72h
Chronic NOEC for Algae / Aquatic Plants	0,3 mg/l

SECTION 12. Ecological information ... / >>

4-MORPHOLINECARBALDEHYDE

LC50 - for Fish	500 mg/l/96h
EC50 - for Crustacea	500 mg/l/48h
EC50 - for Algae / Aquatic Plants	17440 mg/l/72h
EC10 for Algae / Aquatic Plants	10760 mg/l/72h
Chronic NOEC for Crustacea	38 mg/l/21d

LITHIUM CHLORIDE

LC50 - for Fish	158 mg/l/96h oncorhynchus mykiss
EC50 - for Crustacea	> 100 mg/l/48h Daphnia magna
EC50 - for Algae / Aquatic Plants	> 400 mg/l/72h

BUTYL TIN DILAURATE

EC50 - for Algae / Aquatic Plants	1 mg/l/72h

2,2-Dihydroxymethyl butanol

LC50 - for Fish	> 1000 mg/l/96h
EC50 - for Crustacea	13000 mg/l/48h Daphnia magna
EC50 - for Algae / Aquatic Plants	1000 mg/l/72h Pseudokirchnerella subcapitata
Chronic NOEC for Crustacea	> 1000 mg/l Daphnia magna

FATTY ACIDS, TALLOIL, COMPOSED WITH OLEYLAMINE

EC50 - for Algae / Aquatic Plants	7,4 mg/l/72h
Chronic NOEC for Crustacea	4,6 mg/l/21d
Chronic NOEC for Algae / Aquatic Plants	3,1 mg/l

MASS REACTION BETWEEN ETHYLBENZENE AND XYLENE

LC50 - for Fish	> 2,6 mg/l/96h
EC50 - for Crustacea	8,5 mg/l/48h Palaemonetes pugio
EC50 - for Algae / Aquatic Plants	4,6 mg/l/72h
EC10 for Crustacea	1,91 mg/l/21d
EC10 for Algae / Aquatic Plants	> 720 mg/l/73h
Chronic NOEC for Fish	1,3 mg/l/56d
Chronic NOEC for Crustacea	1,57 mg/l/21d
Chronic NOEC for Algae / Aquatic Plants	0,44 mg/l

SOLVENT NAPHTHA (PETROLEUM), LIGHT AROMATIC

LC50 - for Fish	9,2 mg/l/96h Oncorhynchus mykiss
EC50 - for Crustacea	3,2 mg/l/48h Daphnia magna
EC50 - for Algae / Aquatic Plants	2,9 mg/l/72h Pseudokirchneriella subcapitata

UREA-ALDEHYDE RESIN

EC50 - for Crustacea	> 100 mg/l/48h Daphnia magna

XYLENE

LC50 - for Fish	20 mg/l/96h
EC50 - for Crustacea	1 mg/l/48h Daphnia magna
EC50 - for Algae / Aquatic Plants	2,2 mg/l/72h Selenastrum capricornutum
Chronic NOEC for Fish	> 1,3 mg/l/56d Oncorhynchus mykiss
Chronic NOEC for Crustacea	0,96 mg/l/7d Daphnia magna
Chronic NOEC for Algae / Aquatic Plants	0,44 mg/l Pseudokirchneriella subcapitata

TITANIUM DIOXIDE

EC50 - for Crustacea	> 2,41 mg/l/48h
EC50 - for Algae / Aquatic Plants	100 mg/l/72h
EC10 for Algae / Aquatic Plants	2 mg/l/72h
Chronic NOEC for Fish	> 0,004 mg/l/28d
Chronic NOEC for Crustacea	100 mg/l/28d
Chronic NOEC for Algae / Aquatic Plants	1 mg/l/32d

2-METHOXY-1-METHYLETHYL ACETATE

LC50 - for Fish	> 100 mg/l/96h Oncorhynchus mykiss
EC50 - for Crustacea	> 500 mg/l/48h Daphnia magna
EC50 - for Algae / Aquatic Plants	> 1000 mg/l/72h selenastrum capricornutum
Chronic NOEC for Fish	47,5 mg/l/14d Oryzias latipes
Chronic NOEC for Crustacea	100 mg/l/21d Daphnia
Chronic NOEC for Algae / Aquatic Plants	1000 mg/l/72h

SECTION 12. Ecological information ... / >>

ETHYLBENZENE

LC50 - for Fish	> 4,2 mg/l/96h phimephales
EC50 - for Crustacea	1,8 mg/l/48h daphnia magna
EC50 - for Algae / Aquatic Plants	> 4,9 mg/l/72h Pseudokirchneriella subcapitata
Chronic NOEC for Fish	3,3 mg/l/96h
Chronic NOEC for Crustacea	0,96 mg/l/7d
Chronic NOEC for Algae / Aquatic Plants	> 3,4 mg/l/72h

ETHANOL

LC50 - for Fish	14200 mg/l/96h Pimephales promelas
EC50 - for Crustacea	5012 mg/l/48h Ceriodaphnia dubia
EC50 - for Algae / Aquatic Plants	275 mg/l/72h Chlorella vulgaris
Chronic NOEC for Crustacea	9,6 mg/l Daphnia magna

N-BUTYL ACETATE

LC50 - for Fish	18 mg/l/96h Pimephales promelas
EC50 - for Crustacea	44 mg/l/48h Daphnia magna
EC50 - for Algae / Aquatic Plants	675 mg/l/72h Scenedesmus subspicatus
Chronic NOEC for Crustacea	23 mg/l/21d Daphnia magna

12.2. Persistence and degradability

CARBON BLACK
Solubility in water — 1 mg/l @ 20 °C

FATTY ACIDS, C18-UNSATD., TRIMERS, COMPDS. WITH OLEYLAMINE
Solubility in water — 1 mg/l @ 20 °C & pH 6,5
NOT rapidly degradable

4-MORPHOLINECARBALDEHYDE
Solubility in water — 1000 g/l @ 20°C

LITHIUM CHLORIDE
Solubility in water — 569 g/l @ 20°C

BUTYL TIN DILAURATE
Solubility in water — 1,43 mg/l @ 20 °C
NOT rapidly degradable

2,2-Dihydroxymethyl butanol
Solubility in water — 100 g/l @ 25 °C
Entirely degradable

FATTY ACIDS, TALLOIL, COMPOSED WITH OLEYLAMINE
Solubility in water — 1 mg/l @ 20 °C and pH 6 - 6.3
Rapidly degradable

MASS REACTION BETWEEN ETHYLBENZENE AND XYLENE
Solubility in water — 165,8 mg/l @ 25 °C
Rapidly degradable

SOLVENT NAPHTHA (PETROLEUM), LIGHT AROMATIC
Solubility in water — 228,6 mg/l @ 20 °C
Rapidly degradable

XYLENE
Solubility in water — 100 - 1000 mg/l
Rapidly degradable

AMORPHOUS SILICATE HYDRATE
Solubility in water — 0,1 - 100 mg/l
Degradability: information not available

TITANIUM DIOXIDE
Solubility in water — < 0,001 mg/l
Degradability: information not available

SECTION 12. Ecological information ... / >>

BARIUM SULFATE
Solubility in water 0,1 - 100 mg/l
Degradability: information not available

2-METHOXY-1-METHYLETHYL ACETATE
Solubility in water 198 g/l 20°C
Rapidly degradable

ETHYLBENZENE
Solubility in water 1000 - 10000 mg/l
Rapidly degradable

ETHANOL
Solubility in water 1000 - 10000 mg/l
Rapidly degradable

N-BUTYL ACETATE
Solubility in water 5300 mg/l
Rapidly degradable

12.3. Bioaccumulative potential

FATTY ACIDS, C18-UNSATD., TRIMERS, COMPDS. WITH OLEYLAMINE
Partition coefficient: n-octanol/water 5,7 @ 20 °C

4-MORPHOLINECARBALDEHYDE
Partition coefficient: n-octanol/water -1,2 @ 23°C
BCF 1,9 l/kg

LITHIUM CHLORIDE
Partition coefficient: n-octanol/water -1

BUTYL TIN DILAURATE
Partition coefficient: n-octanol/water 4,44 @ 20,8 °C
BCF 2,91

2,2-Dihydroxymethyl butanol
Partition coefficient: n-octanol/water -0,47 @ 26 °C

MASS REACTION BETWEEN ETHYLBENZENE AND XYLENE
Partition coefficient: n-octanol/water 3,16 @ 20 °C
BCF 25,9

SOLVENT NAPHTHA (PETROLEUM), LIGHT AROMATIC
Partition coefficient: n-octanol/water 3,03 @ 20 °C & pH 7

XYLENE
Partition coefficient: n-octanol/water 3,12
BCF 25,9

AMORPHOUS SILICATE HYDRATE
Partition coefficient: n-octanol/water 0,53

2-METHOXY-1-METHYLETHYL ACETATE
Partition coefficient: n-octanol/water 1,2

ETHYLBENZENE
Partition coefficient: n-octanol/water 3,6

ETHANOL
Partition coefficient: n-octanol/water -0,35

N-BUTYL ACETATE
Partition coefficient: n-octanol/water 2,3
BCF 15,3

12.4. Mobility in soil

SECTION 12. Ecological information ... / >>

Information not available

12.5. Results of PBT and vPvB assessment

On the basis of available data, the product does not contain any PBT or vPvB in percentage ≥ than 0,1%.

12.6. Endocrine disrupting properties

Based on the available data, the product does not contain substances listed in the main European lists of potential or suspected endocrine disruptors with environmental effects under evaluation.

12.7. Other adverse effects

Information not available

SECTION 13. Disposal considerations

13.1. Waste treatment methods

Reuse, when possible. Product residues should be considered special hazardous waste. The hazard level of waste containing this product should be evaluated according to applicable regulations.
Disposal must be performed through an authorised waste management firm, in compliance with national and local regulations.
Waste transportation may be subject to ADR restrictions.
CONTAMINATED PACKAGING
Contaminated packaging must be recovered or disposed of in compliance with national waste management regulations.

SECTION 14. Transport information

14.1. UN number or ID number

ADR / RID, IMDG, IATA: UN 1263

14.2. UN proper shipping name

ADR / RID: PAINT
IMDG: PAINT
IATA: PAINT

14.3. Transport hazard class(es)

ADR / RID: Class: 3 Label: 3



IMDG: Class: 3 Label: 3



IATA: Class: 3 Label: 3



14.4. Packing group

ADR / RID, IMDG, IATA: II

14.5. Environmental hazards

ADR / RID: NO
IMDG: NO
IATA: NO

SECTION 14. Transport information ... / >>

14.6. Special precautions for user

ADR / RID:	HIN - Kemler: 33	Limited Quantities: 5 L	Tunnel restriction code: (D/E)
	Special provision: 163, 367, 640D, 650		
IMDG:	EMS: F-E, S-E	Limited Quantities: 5 L	
IATA:	Cargo:	Maximum quantity: 60 L	Packaging instructions: 364
	Passengers:	Maximum quantity: 5 L	Packaging instructions: 353
	Special provision:	A3, A72, A192	

14.7. Maritime transport in bulk according to IMO instruments

Information not relevant

SECTION 15. Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Seveso Category - Directive 2012/18/EU: P5c

Restrictions relating to the product or contained substances pursuant to Annex XVII to EC Regulation 1907/2006

Product
Point 3 - 40
Contained substance
Point 75

Regulation (EU) 2019/1148 - on the marketing and use of explosives precursors
not applicable

Substances in Candidate List (Art. 59 REACH)
On the basis of available data, the product does not contain any SVHC in percentage ≥ than 0,1%.

Substances subject to authorisation (Annex XIV REACH)
None

Substances subject to exportation reporting pursuant to Regulation (EU) 649/2012:
None

Substances subject to the Rotterdam Convention:
None

Substances subject to the Stockholm Convention:
None

Healthcare controls
Workers exposed to this chemical agent must not undergo health checks, provided that available risk-assessment data prove that the risks related to the workers' health and safety are modest and that the 98/24/EC directive is respected.

German regulation on the classification of substances hazardous to water (AwSV, vom 18. April 2017)
WGK 3: Severe hazard to waters

15.2. Chemical safety assessment

A chemical safety assessment has not been performed for the preparation/for the substances indicated in section 3.

SECTION 16. Other information

Text of hazard (H) indications mentioned in section 2-3 of the sheet:

Flam. Liq. 2	Flammable liquid, category 2
Flam. Liq. 3	Flammable liquid, category 3
Muta. 2	Germ cell mutagenicity, category 2
Repr. 1B	Reproductive toxicity, category 1B
Repr. 2	Reproductive toxicity, category 2
Acute Tox. 4	Acute toxicity, category 4
STOT RE 1	Specific target organ toxicity - repeated exposure, category 1
Asp. Tox. 1	Aspiration hazard, category 1
STOT RE 2	Specific target organ toxicity - repeated exposure, category 2
Eye Dam. 1	Serious eye damage, category 1

SECTION 16. Other information ... / >>

Eye Irrit. 2	Eye irritation, category 2
Skin Irrit. 2	Skin irritation, category 2
STOT SE 3	Specific target organ toxicity - single exposure, category 3
Skin Sens. 1	Skin sensitization, category 1
Skin Sens. 1A	Skin sensitization, category 1A
Aquatic Chronic 2	Hazardous to the aquatic environment, chronic toxicity, category 2
Aquatic Chronic 3	Hazardous to the aquatic environment, chronic toxicity, category 3
H225	Highly flammable liquid and vapour.
H226	Flammable liquid and vapour.
H341	Suspected of causing genetic defects.
H360FD	May damage fertility. May damage the unborn child.
H361fd	Suspected of damaging fertility. Suspected of damaging the unborn child.
H302	Harmful if swallowed.
H312	Harmful in contact with skin.
H332	Harmful if inhaled.
H372	Causes damage to organs through prolonged or repeated exposure.
H304	May be fatal if swallowed and enters airways.
H373	May cause damage to organs through prolonged or repeated exposure.
H318	Causes serious eye damage.
H319	Causes serious eye irritation.
H315	Causes skin irritation.
H335	May cause respiratory irritation.
H317	May cause an allergic skin reaction.
H336	May cause drowsiness or dizziness.
H411	Toxic to aquatic life with long lasting effects.
H412	Harmful to aquatic life with long lasting effects.
EUH066	Repeated exposure may cause skin dryness or cracking.

LEGEND:
- ADR: European Agreement concerning the carriage of Dangerous goods by Road
- ATE: Acute Toxicity Estimate
- CAS: Chemical Abstract Service Number
- CE50: Effective concentration (required to induce a 50% effect)
- CE: Identifier in ESIS (European archive of existing substances)
- CLP: Regulation (EC) 1272/2008
- DNEL: Derived No Effect Level
- EmS: Emergency Schedule
- GHS: Globally Harmonized System of classification and labeling of chemicals
- IATA DGR: International Air Transport Association Dangerous Goods Regulation
- IC50: Immobilization Concentration 50%
- IMDG: International Maritime Code for dangerous goods
- IMO: International Maritime Organization
- INDEX: Identifier in Annex VI of CLP
- LC50: Lethal Concentration 50%
- LD50: Lethal dose 50%
- OEL: Occupational Exposure Level
- PBT: Persistent, bioaccumulative and toxic
- PEC: Predicted environmental Concentration
- PEL: Predicted exposure level
- PMT: Persistent, mobile and toxic
- PNEC: Predicted no effect concentration
- REACH: Regulation (EC) 1907/2006
- RID: Regulation concerning the international transport of dangerous goods by train
- TLV: Threshold Limit Value
- TLV CEILING: Concentration that should not be exceeded during any time of occupational exposure.
- TWA: Time-weighted average exposure limit
- TWA STEL: Short-term exposure limit
- VOC: Volatile organic Compounds
- vPvB: Very persistent and very bioaccumulative
- vPvM: Very persistent and very mobile
- WGK: Water hazard classes (German).

GENERAL BIBLIOGRAPHY
1. Regulation (EC) 1907/2006 (REACH) of the European Parliament
2. Regulation (EC) 1272/2008 (CLP) of the European Parliament
3. Regulation (EU) 2020/878 (II Annex of REACH Regulation)
4. Regulation (EC) 790/2009 (I Atp. CLP) of the European Parliament
5. Regulation (EU) 286/2011 (II Atp. CLP) of the European Parliament

SECTION 16. Other information ... / >>

6. Regulation (EU) 618/2012 (III Atp. CLP) of the European Parliament
7. Regulation (EU) 487/2013 (IV Atp. CLP) of the European Parliament
8. Regulation (EU) 944/2013 (V Atp. CLP) of the European Parliament
9. Regulation (EU) 605/2014 (VI Atp. CLP) of the European Parliament
10. Regulation (EU) 2015/1221 (VII Atp. CLP) of the European Parliament
11. Regulation (EU) 2016/918 (VIII Atp. CLP) of the European Parliament
12. Regulation (EU) 2016/1179 (IX Atp. CLP)
13. Regulation (EU) 2017/776 (X Atp. CLP)
14. Regulation (EU) 2018/669 (XI Atp. CLP)
15. Regulation (EU) 2019/521 (XII Atp. CLP)
16. Delegated Regulation (UE) 2018/1480 (XIII Atp. CLP)
17. Regulation (EU) 2019/1148
18. Delegated Regulation (UE) 2020/217 (XIV Atp. CLP)
19. Delegated Regulation (UE) 2020/1182 (XV Atp. CLP)
20. Delegated Regulation (UE) 2021/643 (XVI Atp. CLP)
21. Delegated Regulation (UE) 2021/849 (XVII Atp. CLP)
22. Delegated Regulation (UE) 2022/692 (XVIII Atp. CLP)
23. Delegated Regulation (UE) 2023/707

- The Merck Index. - 10th Edition
- Handling Chemical Safety
- INRS - Fiche Toxicologique (toxicological sheet)
- Patty - Industrial Hygiene and Toxicology
- N.I. Sax - Dangerous properties of Industrial Materials-7, 1989 Edition
- IFA GESTIS website
- ECHA website
- Database of SDS models for chemicals - Ministry of Health and ISS (Istituto Superiore di Sanità) - Italy

Note for users:
The information contained in the present sheet are based on our own knowledge on the date of the last version. Users must verify the suitability and thoroughness of provided information according to each specific use of the product.
This document must not be regarded as a guarantee on any specific product property.
The use of this product is not subject to our direct control; therefore, users must, under their own responsibility, comply with the current health and safety laws and regulations. The producer is relieved from any liability arising from improper uses.
Provide appointed staff with adequate training on how to use chemical products.

CALCULATION METHODS FOR CLASSIFICATION
Chemical and physical hazards: Product classification derives from criteria established by the CLP Regulation, Annex I, Part 2. The data for evaluation of chemical-physical properties are reported in section 9.
Health hazards: Product classification is based on calculation methods as per Annex I of CLP, Part 3, unless determined otherwise in Section 11.
Environmental hazards: Product classification is based on calculation methods as per Annex I of CLP, Part 4, unless determined otherwise in Section 12.

Changes to previous review:
The following sections were modified:
01 / 02 / 03 / 07 / 08 / 09 / 11 / 12 / 14 / 16.